

Predicting Heart Disease Risk from Behavioral and Health Indicators

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🐙 GitHub: <https://github.com/YichenLi95/DATA1030FinalProject>

I. INTRODUCTION

A. Motivation

Heart disease remains one of the leading causes of death in the United States, according to the CDC, One person dies every 34 seconds from cardiovascular disease, and In 2023, in every 3 deaths, one of them is died from cardiovascular disease [1]. Therefore, early detection plays a crucial role. An accurate prediction machine learning model can help identify individuals at higher risk and support timely medical intervention. This project aims to build a machine learning model that predicts whether an individual has heart disease using a combination of health indicators and lifestyle factors.

B. Data Description

The dataset used in this project is from a public Kaggle source [2], contains with synthetic patient records. The dataset contains 253,680 observations and 22 features, including demographic attributes like age, sex, education, income; medical risk factors like BMI, blood pressure, cholesterol; and behavioral indicators like smoking, exercise, fruit/vegetable intake. The target variable, **HeartDiseaseorAttack**, is binary and indicates if an individual has experienced heart disease. Because the dataset is large and each feature has no missing values, it provides a strong foundation for training and evaluating ML models.

C. Previous Work

One publicly available implementation from Kaggle [3] trains a binary logistic regression classifier using a minimal preprocessing pipeline that applies a straightforward train–test split and, only scales BMI using standard scaler, and no hyper-tuning for model training. Their reported performance shows an F1 score of approximately **0.1515**, reflecting their pipelines’ limitations when handling highly imbalanced datasets. In comparison, even though I evaluated my final model using primarily F2 score, my F1 score was **0.3721**, noticeably higher than the previous implementation. This improvement likely results from a more robust pipeline, and a set of models beyond logistic regression, systematic hyperparameter tuning, and cross-validated performance estimation. These methodological enhancements allow the model to better capture complex patterns that a single linear classifier.

II. EXPLORATORY DATA ANALYSIS

During the exploratory data analysis, the most immediate observation was the strong class imbalance in the dataset:

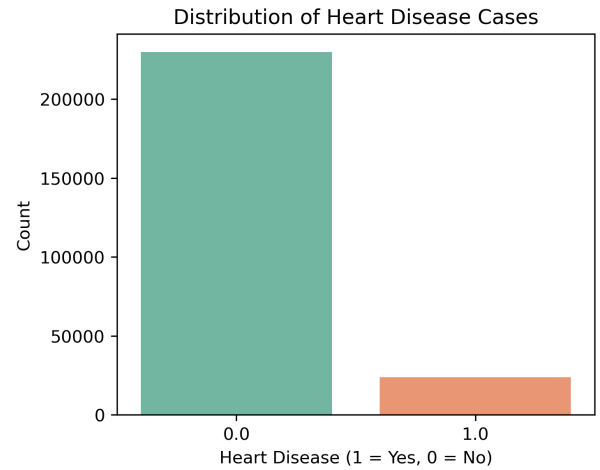


Fig. 1: Heart disease distribution

As shown in Fig 1, only about 9% of respondents reported heart disease. This pattern introduces challenges during model training because most algorithms tend to favor the majority class. This imbalance motivated the later use of stratified splitting to ensure that all training and test sets maintain the same proportion of positive and negative cases.

This project also examined demographic trends. Fig 2 shows that females have a slightly higher rate of heart disease than males. Additionally, the overall population of males is greater than females.

Heart disease distribution by age, Fig 3, is grouped into 13 ordered age bins, from 18 to over 80, the proportion of individuals with heart disease increases steadily from early adulthood into older age groups. The rise is not only monotonic but becomes noticeably sharper after middle age, suggesting that age will likely be one of the top important features in model training.

BMI, one of the continuous health indicators, is also analyzed in Fig 4. The BMI distributions for individuals with and without heart disease share a similar right skewed curve, and their peak is at a similar value around 26. The plot suggests that a scaling method is needed.

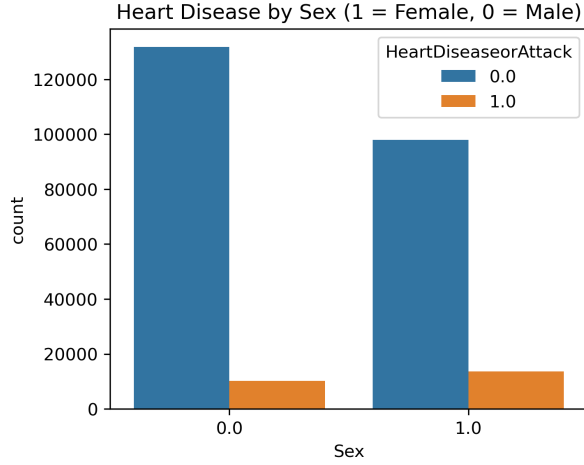


Fig. 2: Heart disease distribution by sex

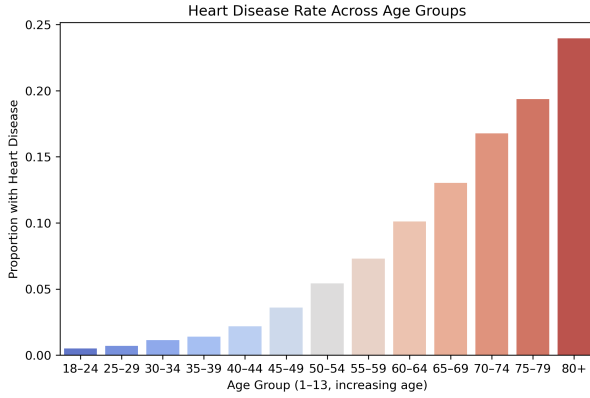


Fig. 3: Heart disease distribution by age

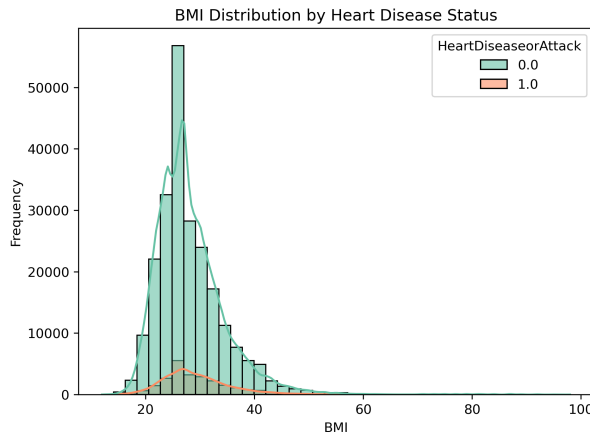


Fig. 4: BMI distribution by Heart Disease Status

Overall, the EDA part revealed three important insights for the modeling pipeline: the need to handle class imbalance carefully, the possible important features, and the importance of feature scaling for continuous variables. These observations directly informed the design choices in the preprocessing and modeling stages that will be introduced later.

III. METHODS

A. Preprocessing

As aforementioned, the dataset is highly imbalanced. Because of this imbalance, a standard random split could easily create subsets that will destroy model training and evaluation. To resolve this issue, an 85/15 stratified train-test split is performed, ensuring that both the training and test sets retain the same ratio of positive to negative cases as the full dataset. (The validation set will be created during model training part.)

Then, a feature transformation process is considered and implemented. The data set did not contain missing values, as this is a synthetic data set. The 21 independent variables fall into three categories: binary, ordinal, and continuous numerical features. 13 of the features are binary indicators, which were already written as 0 or 1 in the original dataset, so no further transformation is required. In addition, ordinal variables - **GENHLTH**, **DIABETE**, **Age**, **Education**, **Income** - were already ordinal encoded, so no additional encoding was applied. At last, the three continuous numerical variables, **BMI**, **MentHlth**, and **PhysHlth** were examined in Fig 5, Fig 6, and Fig 7. As shown in three graphs, BMI values fall between 12 and 98, while both MentHlth and PhysHlth range from 0 to 30 days. Although their distributions are right-skewed, each feature has a clearly defined lower and upper boundary. Therefore, a MinMaxScaler was used to rescale each feature. This choice ensures that all three variables operate on a consistent scale during model training, preventing features with larger numeric ranges from dominating the learning process.

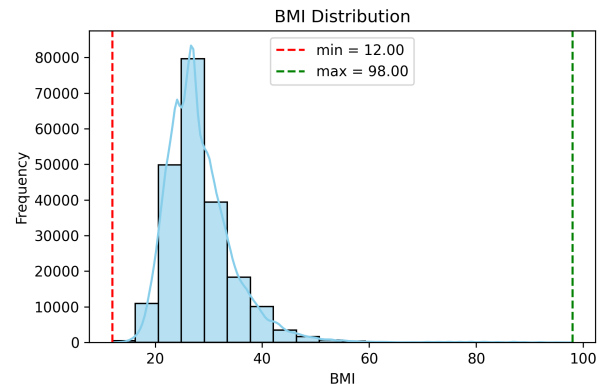


Fig. 5: BMI_distribution

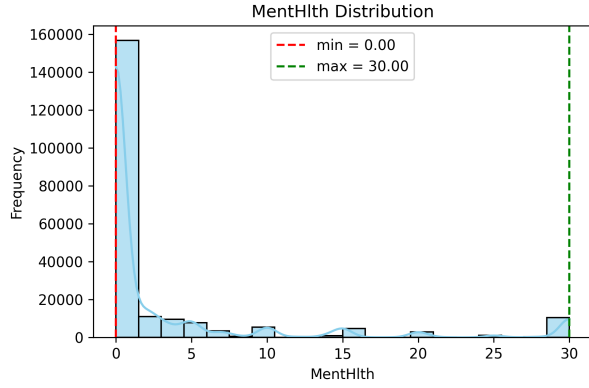


Fig. 6: MentHlth_distribution

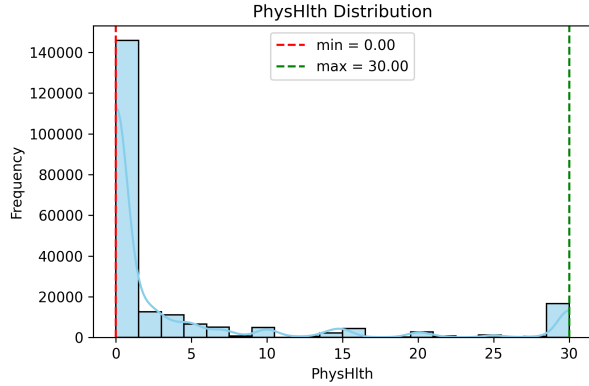


Fig. 7: PhysHlth_distribution

B. ML Pipeline

After preprocessing, a supervised learning pipeline is built. A 5-fold stratified split cross-validation is applied to obtain stable estimates of model performance. Cross-validation reduces the dependence on a single random train-test split and gives a distribution of scores, which I later used to quantify uncertainty.

Four supervised learning algorithms covered in class, Logistic Regression, Random Forest, XGBoost, and K-Nearest Neighbors (KNN), are implemented. These models were selected to ensure diversity in both model assumptions and complexity. Logistic Regression represents a linear classifier. Random Forest introduces nonlinear decision boundaries, while XGBoost extends this with gradient boosting and strong regularization. KNN contributes a model with no explicit training phase, instead relying on geometric structure in the data.

A summary of the tuning ranges is shown in Table I. Logistic Regression mainly tuned the regularization strength C and the penalty type, and used a fix solver, "liblinear". Random Forest varied tree depth and the minimum number of samples required to split a node while used a fixed $n_estimator$ of 500. XGBoost explored learning rate, tree depth, sub-sampling, and column sampling ratio. For KNN, the number of neighbors and

the weighting scheme are tuned.

TABLE I: Hyperparameter Grids Used for Cross-Validation

ML Model	Parameters Tuned
Logistic Regression	$C = [0.01, 0.1, 1, 5, 10]$ penalty = [l1, l2] solver = "liblinear"
Random Forest	max_depth = [1, 5, 10, 30] min_samples_split = [2, 5] n_estimators = 500
XGBoost	learning_rate = [0.01, 0.05, 0.1] max_depth = [1, 3, 5, 8, 10] subsample = [0.3, 0.5, 0.7] colsample_bytree = [0.3, 0.5, 0.7] eval_metric="aucpr"
KNN	n_neighbors = [3, 5, 7] weights = [uniform, distance]

To evaluate model performance, F2 score was used as the primary metric. Since the data is imbalanced, some evaluation metrics, such as accuracy, are not preferred. F2 score weights recall more than precision, which aligns with the practical goal of medical heart disease prediction, where missing a positive case is more harmful than incorrectly predicting disease. For uncertainty measurement, the mean and standard deviation of F2 across cross-validation folds were computed. Models with inherent randomness, such as Random Forest and XGBoost, were evaluated under the same random seed to ensure this pipeline can be re-implemented with no differences.

IV. RESULTS

A. Baseline and Model Performance Comparison

The baseline predicts the majority class for every observation, which means it always outputs "no heart disease." Therefore, according to F2 score's formula, the baseline model for this project is 0.

$$F_2 = \frac{(1 + 2^2) \cdot (\text{Precision} \cdot \text{Recall})}{(2^2 \cdot \text{Precision}) + \text{Recall}}$$

After evaluating these four machine learning models using 5-fold cross-validation and F2 as the primary metric. Table ?? and Fig 8 summarizes the mean and standard deviation of each model's F2 score as well as the best parameter for each model on the validation fold. Among all the models, XGBoost achieved the highest predictive performance at 0.5514, outperforming the baseline by a large margin and achieving several standard deviations above the baseline F2 score. Random Forest and linear regression also showed strong performance but with slightly lower F2 score across folds. KNN performed the weakest among the non-linear models, likely because distance-based methods struggle with high-dimensional data.

TABLE II: Model Performance Comparison and Best Param

ML Model	Best Parameters	Mean F2	Std F2
Logistic Regression	C = 1.0 penalty = l1 solver = liblinear	0.5493	0.00335
Random Forest	max_depth = 10 min_samples_split = 2	0.5461	0.00235
XGBoost	colsample_bytree = 0.7 learning_rate = 0.05 max_depth = 5 subsample = 0.5	0.5514	0.00346
KNN	n_neighbors = 3 weights = distance	0.2091	0.00283

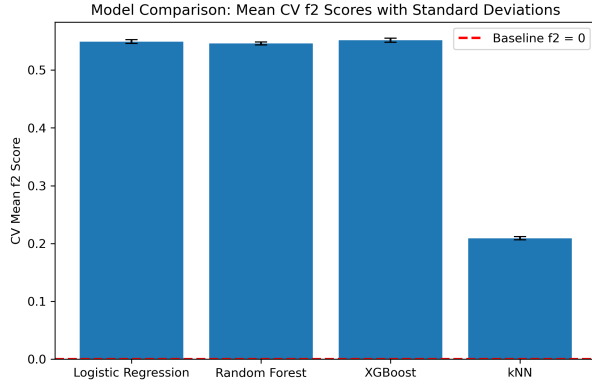


Fig. 8: Model Performance Comparison

A confusion matrix, see Fig 9, is computed on the test set.

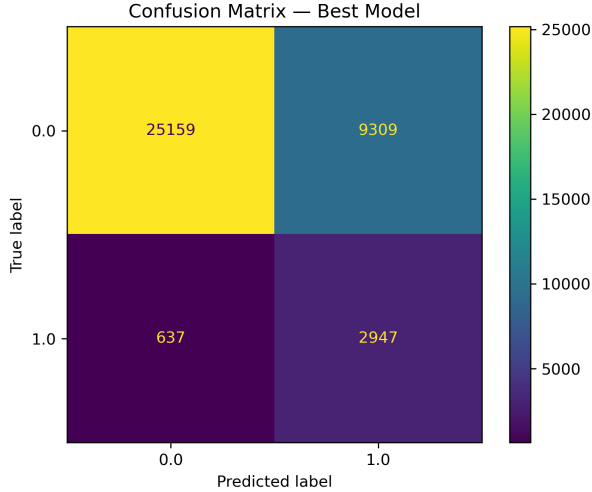


Fig. 9: Confusion Matrix

The model correctly identified 25,159 true negatives and 2,947 true positives, while producing 9,309 false positives and 637 false negatives. These results show that the model successfully captures a substantial portion of the actual heart disease cases. The model achieved an F2 score of 0.5541, which aligns well with the cross-validation results. The relatively low number of false negative cases is encouraging as in the medical-risk setting, missing a positive case is worse than issuing a false alarm.

B. Feature Importance

To better understand what drives the model's predictions, three different XGBoost global feature-importance metrics, gain, weight, and cover, are computed. As shown in Fig 10, 11, and 12, even each metric measures importance differently, all three produce a highly consistent ranking of predictive features.

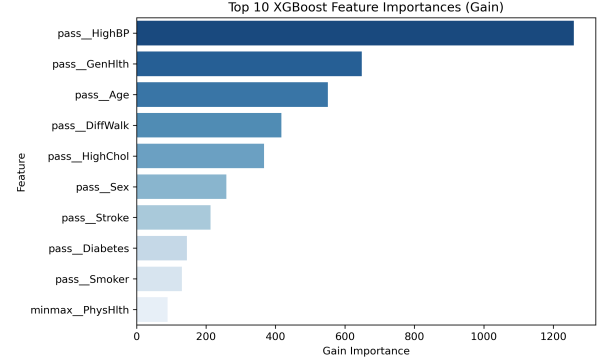


Fig. 10: Gain Metric

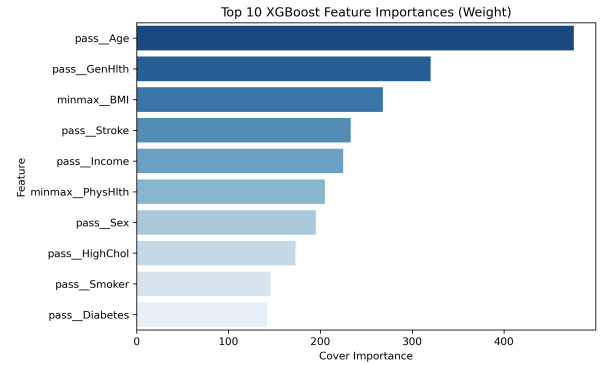


Fig. 11: Weight Metric

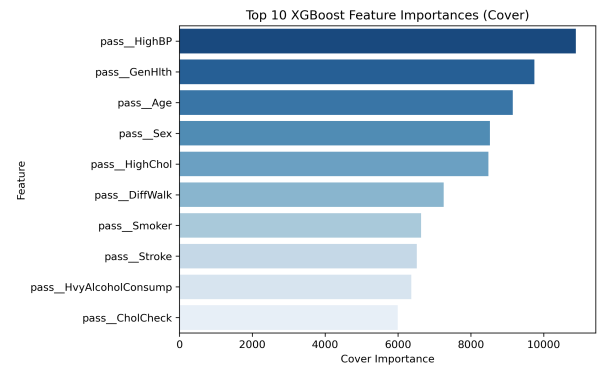


Fig. 12: Cover Metric

Across these metrics, High Blood Pressure (**HighBP**), General Health(**GenHlth**), and **Age** stand as the strongest predictors of heart disease. These variables either provide

the most information gain when splitting nodes, appear most frequently across the trees, or account for the largest share of samples covered by important splits. Other lifestyle-related indicators, including High Cholesterol(**HighChol**) and difficulty walking(**DiffWalk**), also rank highly across metrics. The agreement among these metrics strengthens the point that the model is capturing meaningful patterns.

Fig 13 and 14 show the generated SHAP force plots for two indices observations: index 0 and index 20,000. These two plots show how each feature pushes the prediction above or below the baseline risk.

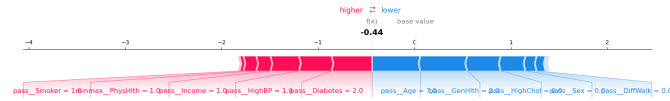


Fig. 13: Force Plot at index 0

For index 0, the model predicts a SHAP value of -0.44. The force plot indicates that **Age**, General Health(**GenHlth**), and High Cholesterol(**HighChol**) pull the score downward. Meanwhile, High Blood Pressure(**HighBP**), **Diabetes**, and Low **Income** push it upward. Overall the probability of this individual of getting heart attack is below the baseline.

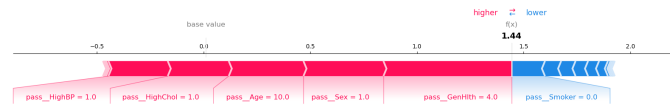


Fig. 14: Force Plot at index 20000

For index 20,000, the opposite pattern appears. The predicted value of 1.44 is above the baseline, driven mainly by High Blood Pressure(**HighBP**), High Cholesterol(**HighChol**), and Older **Age**. These features have strong positive contributions. The only notable negative contribution is not smoking(**Smoker**), but this effect is small compared to the dominant risk-enhancing factors. Therefore, this point is assigned a substantially higher probability of heart disease.

To conclude, these examples demonstrate that the model knows which features tend to increase cardiovascular risk and vice versa.

V. OUTLOOK

There are two meaningful directions to further improve this project.

First, the hyperparameter search space can be expanded. For instance, KNN performed relatively poorly, likely because its parameter grid was too narrow. Exploring a wider range of neighbors, distance metrics, or even replacing KNN with a more robust distance-based classifier could lead to significant gains. Similarly broadening the search for Random Forest and XGBoost may help extract stronger predictive patterns from

the data. Due to computation limits, these thoughts are not achieved in this project.

Additionally, model performance could be benefited from merging additional informative features. Important clinical indicators, such as family history of cardiovascular disease or numerical blood pressure measurements, would likely increase predictive power. Adding additional features may also improve interpretability.

ACKNOWLEDGMENTS

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REFERENCES

- [1] Heart Disease Facts. (2024, October 24). Heart Disease. <https://www.cdc.gov/heart-disease/data-research/facts-stats/index.html>
- [2] Heart Disease Health Indicators Dataset. (2022, March 10). Kaggle. <https://www.kaggle.com/datasets/alexteboul/heart-disease-health-indicators-dataset>
- [3] Binary Logistic Regression. (2023, June 28). Kaggle. <https://www.kaggle.com/code/minthiatun/binary-logistic-regression/notebook>